SWEN 221

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Inner classes

Static Inner classes
Hierarchical code organization

Static Inner classes, often called nested classes in other languages, are just a way to do Hierarchical code organization.

Their type is simply a composed type, like `Exp.BinOp` or `Exp.BinOp.Op`

The fact that the static class `BinOp` is inside `Exp` have no operational semantic.

(Non-static) Inner classes, are much more complex, that Static Inner classes!
Note: inner records are implicitly static, and may be better for this use case!

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public interface Exp {

   ...
}
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  }

  ...
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  public static class FieldAccess implements Exp{
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  }
  public static class MethodCall implements Exp{
    Exp receiver; String mName; List<Exp> parameters;
  }



  ...
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```

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  }
  public static class MethodCall implements Exp{
    Exp receiver; String mName; List<Exp> parameters;
  }
  public static class BinOp implements Exp{
    Op op; Exp left; Exp right;

  }
  ...
}
```

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  public static class MethodCall implements Exp{
    Exp receiver; String mName; List<Exp> parameters;
  }
  public static class BinOp implements Exp{
    Op op; Exp left; Exp right;
    public static enum Op{ Plus, Minus, And, Or, ... }
  }
  ...
}
```

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Static Inner classes
Hierarchical code organization

```
class Foo {
  public static class Bar {...}
  private static class Beer {...}
  ...
}
public class Main {
  public static void main(String[] args){
    Foo.Bar bar= new Foo.Bar();
    System.out.println(bar);
    System.out.println(Exp.BinOp.Op.Plus);
  }
}
```

(Non-static) Inner Classes

- static inner classes → property of classes
  - a single `Foo.Bar` class exists
- inner classes → property of instances
  - each instance have its own (non-static) inner classes

- In the same way as

  static fields → property of classes

  fields → property of instances

Inner Classes: Example

```
class IntList implements Iterable<Integer> {
  private int[] data;
  private int size= 0;
  /* ... */
  public IntList() {this.data= new int[4];}
  public Iterator<Integer> iterator() {
    return new InternalIter();
  }
  private class InternalIter implements Iterator<Integer>{
    private int pos= 0;
    public boolean hasNext() {return pos < size;}
    public Integer next(){return data[pos++];}
    /* ... */
  }
}
```

Inner Class

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    public boolean hasNext() {return pos < size;}
    public Integer next(){return data[pos++];}
    /* ... */
  }
}
```

Can access private fields/methods of enclosing class

Inner Classes: Example

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  private int[] data;
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  /* ... */
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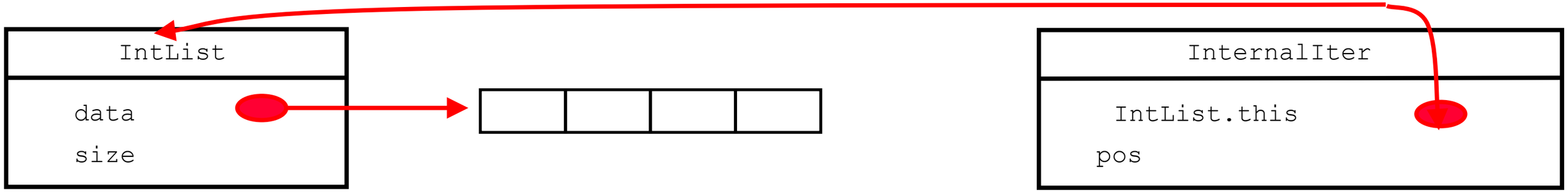
  }
  private class InternalIter implements Iterator<Integer>{
    private int pos= 0;
    public boolean hasNext() {return pos < size;}
    public Integer next(){return data[pos++];}
    /* ... */
  }
}
```

Enclosing class can construct and return instances of inner class

Other classes cannot construct instances as it's private

Inner Classes: Scoping

- Inner classes have *outer pointer*
  - For accessing fields/methods of enclosing class (outer)
  - Outer pointer automatically supplied for new inner class




```
class IntList implements Iterable<Integer> {
  private int[] data;
  private int size= 0;/* ... */
  private class InternalIter implements Iterator<Integer>{
    private int pos= 0;/* ... */
  }
}
```

Inner Classes: Explicit Scoping

```
class IntList implements Iterable<Integer> {
  private int[] data;
  private int size= 0;
  /* ... */
  }
  private class InternalIter implements Iterator<Integer>{
    private int pos= 0;
    public Integer next(){
      return IntList.this.data[InternalIter.this.pos++];
    }
    /* ... */
  }
}
```

This line is now fully explicit in this-scoping

Inner Classes: Explicit Scoping

```
class IntList implements Iterable<Integer> {
  private int[] data;
  private int size= 0;
  /* ... */
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  private class InternalIter implements Iterator<Integer>{
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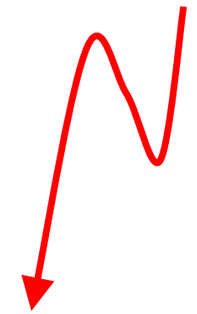
    public Integer next(){
      return this.data[IntList.this.pos++];
    }
    /* ... */
  }
}
```



Wrong explicit scoping here

Inner Classes: Explicit Scoping

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  /* ... */
  }
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    private int pos= 0;
    public Integer next(){
      return InternalIter.this.data[this.pos++];
    }
    /* ... */
  }
}
```

Wrong explicit scoping here

Inner Classes: External Construction

```
class Shape {
  /* ... */
  public class Square {
    private int x, y, width, height;
    public Square(int x, int y, int width, int height){
      /* ... */
    }
  }
}
    /* ... */
    Shape outer= new Shape();
    Shape.Square square= outer.new Square(0,0,8,42);
    square= new Shape().new Square(0,0,8,42);
```

- External Construction
  - If constructing inner class outside outer, or in static method, must supply outer pointer **explicitly**

Inner Classes - Static Inner Classes

- Static Inner Classes have no outer pointer!
  - So, can not access fields/methods of enclosing class
  - But, can construct without providing outer pointer
  - If no need to access enclosing info, then this is more convenient (and potentially more efficient)

Inner Classes - Static Inner Classes

- Static Inner Classes have no outer pointer!
  - So, can not access fields/methods of enclosing class
  - But, can construct without providing outer pointer
  - If no need to access enclosing info, then this is more convenient (and potentially more efficient)
- (Non-Static) Inner Classes have outer pointer!
  - So, they have multiple `this` and can use it to (implicitly/explicitly) access fields/methods of enclosing instance
  - But, can not be instantiated without providing the outer pointer

Method Local Inner Classes

```
class Outer {
  public Outer create(final int field) {
    class Inner extends Outer {
      private int myfield= field;
      /*...*/
    };

    return new Inner();
  }
}
```

Non-static method local class

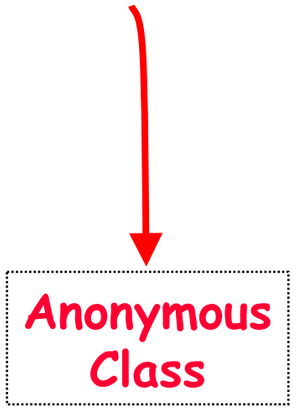
Can access local variables + parameters provided they are (effectively) final.

- Can even define classes within a method!
  - These are only visible within that method
  - But, their instances can still be returned
  - Cannot have static method-local classes

Anonymous Classes: Example

```
public static void main(String[] args) {
  List<String> myList= new ArrayList<String>(){
    // override ArrayList.add
    public boolean add(String x) {
      System.out.println("ADDED: " + x);
      return super.add(x);
    }
  };
}
```

- Anonymous Class
  - Has no class definition and, hence, no name
  - Defined as an extension of existing class
  - Can override methods and/or define fields



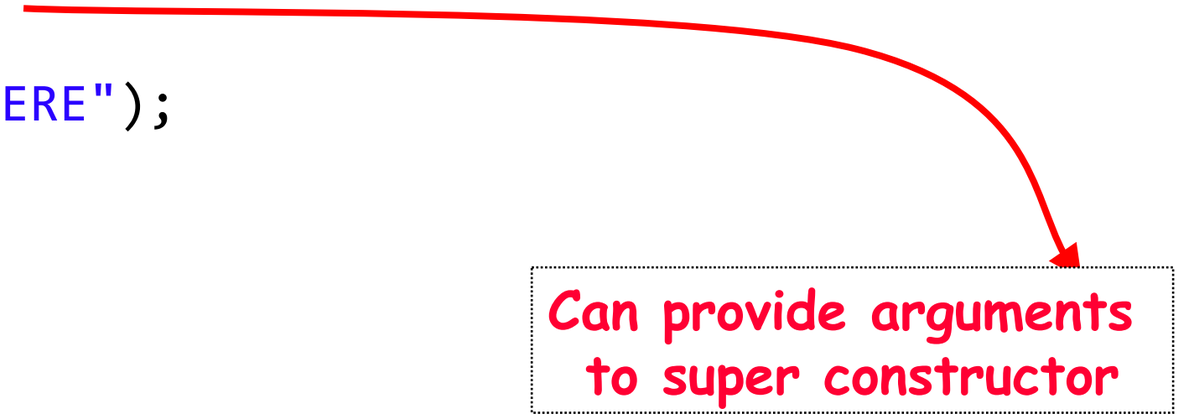
Anonymous Classes: Syntax

```
public class Test {
  private int field;
  public Test(int field){ this.field= field; }
  public void aMethod(){ /*...*/ }

  public static void main(String[] a){
    Test x= new Test() {
      public void aMethod() {
        System.out.println("GOT HERE");
      }
    };
  }
}
```

Compile time error

Anonymous Classes: Syntax

```
public class Test {
  private int field;
  public Test(int field){ this.field= field; }
  public void aMethod(){ /*...*/ }

  public static void main(String[] a){
    Test x= new Test(1) {
      public void aMethod() {
        System.out.println("GOT HERE");
      }
    };
  }
}
```



Can provide arguments to super constructor

Anonymous Classes: Syntax

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public class Test {
  private int field;
  public Test(int field){ this.field= field; }
  public void aMethod(){ /*...*/ }

  public static void main(String[] a){
    Test x= new Test(1) {
      public void aMethod() {
        System.out.println("GOT "+a+" "+field);
      }
    };
  }
}
```

Can access local variables and parameters
(provided they are effectively final)
and enclosing fields

Anonymous Classes: Interfaces

```
ArrayList<String> ls= /*...*/;
Collections.sort(ls, new Comparator<String>(){
  public int compare(String s1, String s2) {
    return s1.compareToIgnoreCase(s2);
    }});
```

Can even make anonymous class from an interface!!!

Very common case: interface with just 1 method.
Lambdas are just a syntactic sugar for anonymous classes

Anonymous Classes: Interfaces

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ArrayList<String> ls= /*...*/;
Collections.sort(ls, new Comparator<String>(){
  public int compare(String s1, String s2) {
    return s1.compareToIgnoreCase(s2);
    }});
```

- Learn how to read the code through the syntax:
  - fading away the anonymous class+method declaration, what you obtain reads like:

Sort `ls` using `s1.compareToIgnoreCase(s2)`

Anonymous Classes: Interfaces

```
ArrayList<String> ls= /*...*/;
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  public int compare(String s1, String s2) {
    return s1.compareToIgnoreCase(s2);
    }});
```

- Learn how to read the code through the syntax:
  - fading in the opposite way, we see type information they double check that we know what we are doing

```
int Comparator<String>.compare(String,String)
```

Before Java8: Used a lot for event handlers

```
import java.awt.*; import java.awt.event.*; import javax.swing.*;

public class MiniGui extends JFrame {
  public static void main(String[] args) {
    SwingUtilities.invokeLater(new Runnable() {
      public void run() {
        MiniGui g= new MiniGui();
        g.setDefaultCloseOperation(WindowConstants.EXIT_ON_CLOSE);
        g.getRootPane().setLayout(new BorderLayout());
        JButton b= new JButton("------Bar------");
        b.addActionListener(new ActionListener() {
          public void actionPerformed(ActionEvent e) {
            System.out.println("Button pressed");
          }});
        g.getRootPane().add(b, BorderLayout.CENTER);
        g.pack();
        g.setVisible(true);
      }});
  }}
```

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After Java8: Lambdas for event handlers

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      });
  }
}
```

Example: Groups of Persons

-A Person have a String name and many Person friends.
-A Group of friend represent a set of friendships.

We want to enforce the following:
-A Person only belong to a single Group, and have friends only in that Group.

```
public class Group{
  private List<Person> ps= new ArrayList<>();
  public List<Person> persons(){ return Collections.unmodifiableList(ps); }
  public void addPerson(String name){ ps.add(new Person(name)); }

  private void checkInGroup(Person p){ if (!ps.contains(p)){ throw new Error(""); } }
  public void connect(Person p1, Person p2){
    checkInGroup(p1);
    checkInGroup(p2);
    if (p1 == p2){ return; }
    if (p1.friends.contains(p2)){ return; }
    p1.friends.add(p2);
    p2.friends.add(p1);
  }
  public final static class Person{//important: nested
    private String name;//private mutable state
    public String name(){ return this.name; }//get, but no setter
    private Person(String name){ this.name= name; }//private constructor

    private List<Person> friends= new ArrayList<>();//private mutable state
    public List<Person> friends(){ return Collections.unmodifiableList(friends); }
    public String toString(){ return ...; }
  }
}
```

Private means: private to the whole set of nested classes

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    checkInGroup(p2);
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    if (p1.friends.contains(p2)){ return; }
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    p2.friends.add(p1);
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    public List<Person> friends(){ return Collections.unmodifiableList(friends); }
    public String toString(){ return ...; }
  }
}
```

Private means: private to the whole set of nested classes

Groups of Persons

How is up to now?

- Success?
  - Can the user modify Person directly?
    - no, all mutation operation are private, so **only Groups can modify them**.
  - **code logic** check that persons of different groups are never connected.
  - Persons are **created by the Group**, and not directly by the user.
- A delicate balance! Change one thing, all collapse!

A more usable Group

How to expose only a read view of a group:
ReadGroup is an interface, Group will implement it, and there will be a method to wrap a Group in a ReadGroup.
Note how there is **no way** to extract the inner group from the result of a ReadGroup.of(..)

```
public class Group implements ReadGroup{...}

public interface ReadGroup{
  public List<Group.Person> persons();
  public static ReadGroup of(Group g){
    return new ReadGroup(){
      public List<Group.Person> persons(){
        return g.persons();
      }
    };
  }
}
```

A more usable Group

- How offer a deepClone() operation:
  Hard to do by hand: circular graph of friends!
  Java idiomatic way: use serialization!

```
public class Group implements ReadGroup,Serializable{...
  public final static class Person implements Serializable{..}
  public Group deepClone(){ return DeepClone.copy(this); }
}

class DeepClone{
  @SuppressWarnings("unchecked")
  public static <T> T copy(T orig){
    var aux= new ByteArrayOutputStream();
    try (var o= new ObjectOutputStream(aux)){ o.writeObject(orig); o.flush(); }
    catch(IOException e){ throw new Error(e); }
    var a= aux.toByteArray();
    try (var i= new ObjectInputStream(new ByteArrayInputStream(a))){
      return (T)i.readObject();
    }
    catch(IOException | ClassNotFoundException e){ throw new Error(e); }
  }
}
```

Group: testing/example usage

```
String s(ReadGroup g){ return g.persons().toString(); }

@Test public void test() {
  Group g = new Group();
  g.addPerson("Bob");//persons created by the group
  g.addPerson("Alice");

  //modification handled by the group
  g.connect(g.persons().get(0), g.persons().get(1));
  assertEquals("[Bob[Alice], Alice[Bob]]",s(g));

  ReadGroup wrapper=ReadGroup.of(g);//read only view
  ReadGroup imm=ReadGroup.of(g.deepClone());//immutable datatype
  assertEquals("[Bob[Alice], Alice[Bob]]",s(wrapper));
  assertEquals("[Bob[Alice], Alice[Bob]]",s(imm));

  g.addPerson("Wally");//modifies only wrapper
  assertEquals("[Bob[Alice], Alice[Bob], Wally[]]",s(wrapper));
  assertEquals("[Bob[Alice], Alice[Bob]]",s(imm));
}
```

Mutable and Immutable Groups

- Now we can create Groups as mutable, mutate them as long as we want, and then turn them immutable by doing
  `"ReadGroup.of(g.deepClone())"`

- Thus, following those patterns, in Java we can define mutable and immutable data and make them play together.

Next time – equals is hard!